

# TASK B - Diagnose a slow site

## TASK LIST: -

1. Website detail what we use
2. Website Initial Performance Diagnosis
3. Problems what we & Their Impact in website
4. Prioritized Fixes
5. What we changed to do better /before vs after test (made a demo of our own)
6. Client-Facing Summary & Conclusion

## 1. Website detail what we use

**Website:** BioCollex Research

**URL:** <https://biocollexresearch.com/>

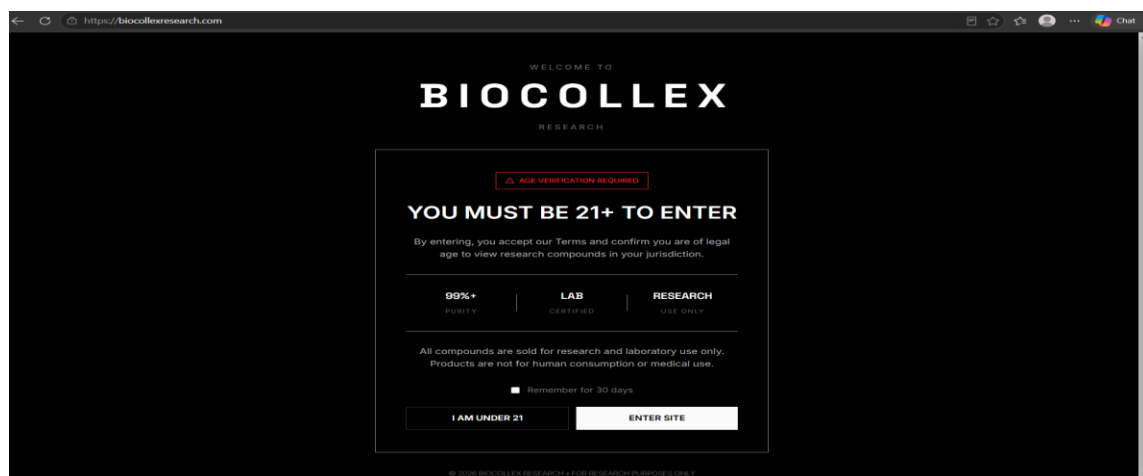
**Test:** Mobile performance analysis

**Tool:** Google PageSpeed Insights / Lighthouse

**Test Date:** 6 August 2026

BioCollex Research was selected as the test website because it is a real, publicly accessible website and showed significant performance issues on mobile during testing.

The analysis focused on identifying the main causes of slow loading, understanding their impact on users, and testing a lightweight rebuilt section to demonstrate how performance could be improved.



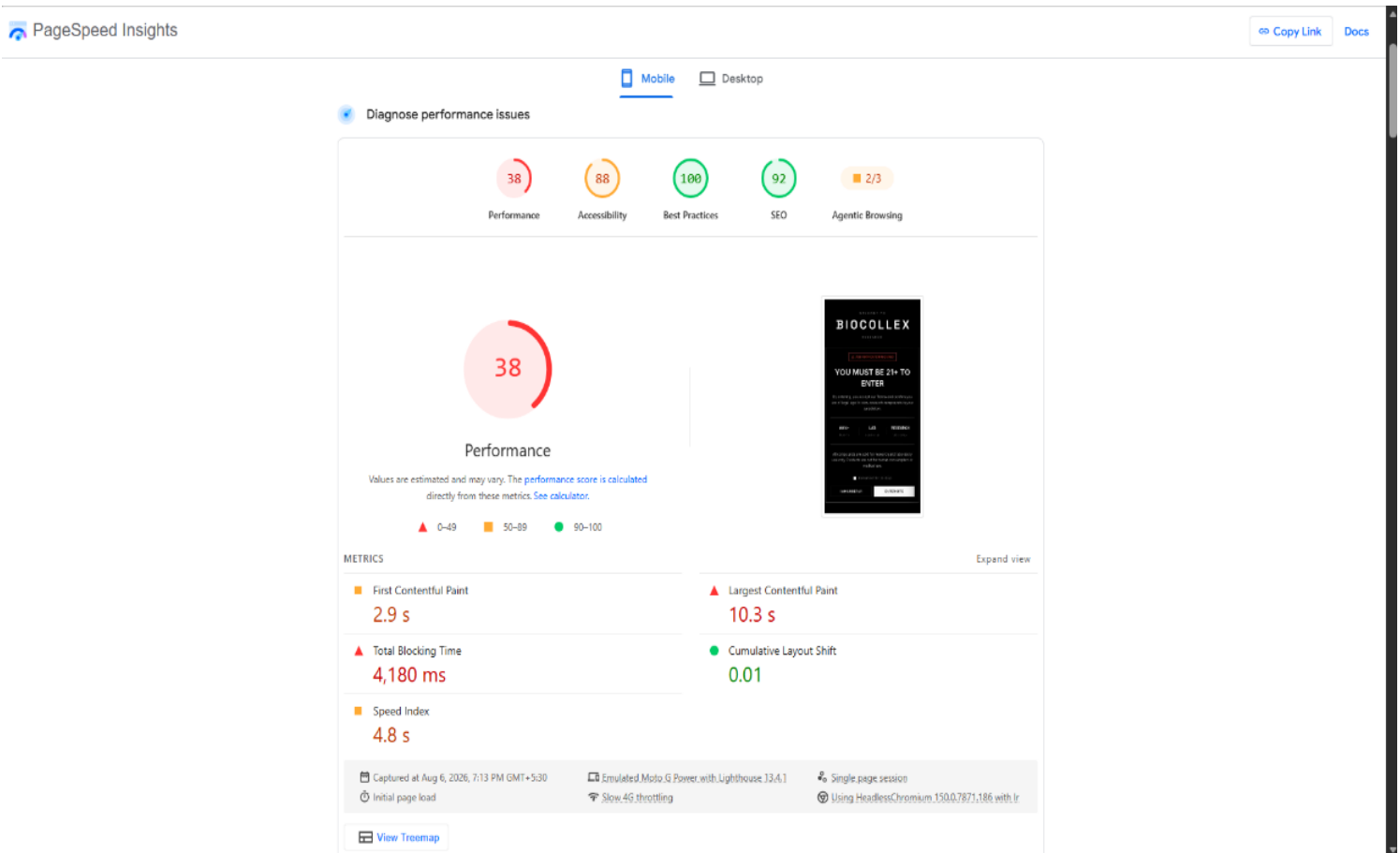
## 2. Website Initial Performance Diagnosis

The initial mobile Lighthouse test gave the website a **38/100 Performance score**, indicating significant performance issues.

| Metric             | Result   | Observation                          |
|--------------------|----------|--------------------------------------|
| <b>Performance</b> | 38/100   | Poor mobile performance              |
| <b>FCP</b>         | 2.9 s    | First content appeared slowly        |
| <b>LCP</b>         | 10.3 s   | Main content took too long to appear |
| <b>TBT</b>         | 4,180 ms | Heavy JavaScript/main-thread work    |
| <b>CLS</b>         | 0.01     | Layout stability was good            |
| <b>Speed Index</b> | 4.8 s    | Visual loading was relatively slow   |

The most important problems were the **10.3-second LCP** and **4,180 ms TBT**. These indicate that **users on mobile may have to wait several seconds** before the main content becomes visible and the page becomes fully responsive.

The test also showed a **total network payload of approximately 6.4 MB**, making the page unnecessarily heavy for mobile users.



### 3. Problems Found & Impact

The diagnosis identified five main performance problems:

| Problem Found       | Evidence                              | Impact                                                       |
|---------------------|---------------------------------------|--------------------------------------------------------------|
| Large hero video    | ~5 MB video file                      | Increased download time and delayed the main content         |
| Heavy JavaScript    | ~4.5 s JavaScript execution           | Increased processing time and reduced responsiveness         |
| Large page weight   | ~6.4 MB total payload                 | Slow loading on mobile and slower networks                   |
| Render-blocking CSS | ~240 KB CSS, ~1.95 s estimated delay  | Delayed the initial page rendering                           |
| Third-party scripts | Facebook scripts used ~1.9 s CPU time | Added processing that was not essential for the main content |

#### Main Finding

The biggest performance costs came from the **large hero video and excessive browser work from JavaScript and other resources.**

These issues are especially important on mobile because users may have slower networks and less powerful devices. Reducing unnecessary media, scripts and blocking resources should therefore provide a more noticeable improvement than minor visual or code-level optimizations.

#### INSIGHTS

Render-blocking requests — Est savings of 1,200 ms

Requests are blocking the page's initial render, which may delay LCP. [Deferring or inlining](#) can move these network requests out of the critical path. LCP FCP Unscored

Show 3rd-party resources (1)

| URL                                                                      | Transfer Size | Duration |
|--------------------------------------------------------------------------|---------------|----------|
| biocollexresearch.com <span>1st Party</span>                             | 240.5 KiB     | 1,950 ms |
| ...1/f3a7e09....css (biocollexresearch.com)                              | 240.5 KiB     | 1,950 ms |
| Google Fonts <span>Cdn</span>                                            | 1.3 KiB       | 750 ms   |
| /css2?family=Montserrat:wght@400;600&display=swap (fonts.googleapis.com) | 1.3 KiB       | 750 ms   |

#### DIAGNOSTICS

▲

Minimize main-thread work

— 13.8 s

Consider reducing the time spent parsing, compiling and executing JS. You may find delivering smaller JS payloads helps with this. [Learn how to minimize main-thread work](#) TBT Unscored

| Category                     | Time Spent |
|------------------------------|------------|
| Style & Layout               | 4,705 ms   |
| Script Evaluation            | 4,216 ms   |
| Other                        | 2,967 ms   |
| Script Parsing & Compilation | 960 ms     |
| Rendering                    | 356 ms     |
| Parse HTML & CSS             | 312 ms     |
| Garbage Collection           | 248 ms     |

**Reduce JavaScript execution time — 4.5 s**

Consider reducing the time spent parsing, compiling, and executing JS. You may find delivering smaller JS payloads helps with this. [Learn how to reduce Javascript execution time.](#) (TBT) (Unscored)

☒ Show 3rd-party resources (2)

| URL                                                                            | Total CPU Time | Script Evaluation | Script Parse |
|--------------------------------------------------------------------------------|----------------|-------------------|--------------|
| biocollexresearch.com (1st Party)                                              | 9,565 ms       | 2,503 ms          | 365 ms       |
| https://biocollexresearch.com                                                  | 5,797 ms       | 619 ms            | 12 ms        |
| ...jquery/jquery.min.js?ver=3.7.1 (biocollexresearch.com)                      | 880 ms         | 246 ms            | 81 ms        |
| ...js/frontend-modules.min.js?ver=4.1.4 (biocollexresearch.com)                | 799 ms         | 764 ms            | 17 ms        |
| ...js/lenis.min.js?ver=1.1.19 (biocollexresearch.com)                          | 468 ms         | 80 ms             | 13 ms        |
| ...wpms/lenis-init.min.js?ver=176... (biocollexresearch.com)                   | 288 ms         | 19 ms             | 0 ms         |
| ...js/jquery.ripples.js?ver=1.0.0 (biocollexresearch.com)                      | 284 ms         | 275 ms            | 9 ms         |
| ...js/frontend.min.js?ver=4.1.4 (biocollexresearch.com)                        | 215 ms         | 81 ms             | 19 ms        |
| ...1/f3a7e09...css (biocollexresearch.com)                                     | 146 ms         | 0 ms              | 0 ms         |
| ...js/shared-frontend-handlers.03caa53...bundle.min.js (biocollexresearch.com) | 120 ms         | 100 ms            | 20 ms        |
| ...frontend/order-attribution.min.js?ver=10.9.4 (biocollexresearch.com)        | 113 ms         | 112 ms            | 1 ms         |
| ...js/soundjs.min.js?ver=1.0.0 (biocollexresearch.com)                         | 109 ms         | 24 ms             | 85 ms        |
| ...17.6.3/lazyload.min.js (biocollexresearch.com)                              | 105 ms         | 94 ms             | 2 ms         |
| ...js/frontend.min.js?ver=1.7.1064 (biocollexresearch.com)                     | 102 ms         | 7 ms              | 91 ms        |
| ...js/text-editor.4560966...bundle.min.js (biocollexresearch.com)              | 73 ms          | 63 ms             | 1 ms         |
| ...jarallax/jarallax.min.js?ver=1.12.7 (biocollexresearch.com)                 | 65 ms          | 21 ms             | 12 ms        |
| Facebook (Social)                                                              | 1,939 ms       | 1,217 ms          | 282 ms       |
| ...config/173...?v=... (connect.facebook.net)                                  | 1,313 ms       | 808 ms            | 157 ms       |
| /en_US/fbevents.js (connect.facebook.net)                                      | 626 ms         | 409 ms            | 125 ms       |
| Unattributable                                                                 | 1,444 ms       | 58 ms             | 78 ms        |
| Unattributable                                                                 | 1,014 ms       | 16 ms             | 0 ms         |
| about:srcdoc                                                                   | 323 ms         | 28 ms             | 5 ms         |
| data:text/javascript;base64,CihmdW5jdGlvbigpe3ZhciBoZWFKZXUIVE1...             | 107 ms         | 14 ms             | 72 ms        |

**Avoid enormous network payloads — Total size was 6,439 KiB**

Large network payloads cost users real money and are highly correlated with long load times. [Learn how to reduce payload sizes.](#) (Unscored)

 Consider showing excerpts in your post lists (e.g. via the more tag), reducing the number of posts shown on a given page, breaking your long posts into multiple pages, or using a plugin to lazy-load comments.

☒ Show 3rd-party resources (4)

| URL                                                        | Transfer Size |
|------------------------------------------------------------|---------------|
| biocollexresearch.com (1st Party)                          | 5,593.6 KiB   |
| ...04/hero-bg.mp4 (biocollexresearch.com)                  | 5,011.5 KiB   |
| ...1/f3a7e09...css (biocollexresearch.com)                 | 240.5 KiB     |
| ...02/GLP2-TZ-430x430.png (biocollexresearch.com)          | 163.9 KiB     |
| https://biocollexresearch.com                              | 64.7 KiB      |
| ...js/frontend.min.js?ver=1.7.1064 (biocollexresearch.com) | 63.0 KiB      |
| ...03/image-removebg-preview.png (biocollexresearch.com)   | 50.0 KiB      |
| Facebook (Social)                                          | 230.8 KiB     |
| ...config/173...?v=... (connect.facebook.net)              | 121.8 KiB     |
| /en_US/fbevents.js (connect.facebook.net)                  | 109.0 KiB     |
| Google Fonts (Cdn)                                         | 96.2 KiB      |
| ...v20/UcC73FwrK...woff2 (fonts.gstatic.com)               | 48.1 KiB      |
| ...v20/UcC73FwrK...woff2 (fonts.gstatic.com)               | 48.1 KiB      |

## 4. Prioritized Fix List

The fixes were prioritized based on expected performance impact compared with implementation effort.

| Priority | Recommended Fix                                              | Expected Impact | Effort |
|----------|--------------------------------------------------------------|-----------------|--------|
| High     | Replace/compress the ~5 MB hero video                        | Very High       | Low    |
| High     | Reduce unnecessary JavaScript and defer non-critical scripts | High            | Medium |
| High     | Reduce unused CSS and optimize the large stylesheet          | High            | Medium |
| Medium   | Optimize large images and use modern formats                 | Medium          | Low    |
| Medium   | Delay non-essential third-party scripts                      | Medium          | Low    |
| Low      | Optimize fonts and loading behavior                          | Low–Medium      | Low    |

### What We Would Not Prioritize

We would not spend significant development time on minor optimizations that have little effect on the main bottlenecks, such as **small HTML changes** or **tiny reductions** in already-small files.

The priority should remain on the hero **media, JavaScript execution, CSS** delivery and overall page weight, because these have the greatest potential impact on mobile performance.

## 5. Rebuilt Section Demo/ Before vs After

To validate the recommendation, the website's hero section was rebuilt as a lightweight standalone demo.

The original page **used a large background video of approximately 5 MB**, which contributed **significantly to the page's 6.4 MB total network payload**.

For the demo, the heavy video-based hero was replaced with a lightweight, responsive HTML/CSS section containing only the essential content and a clear call-to-action.

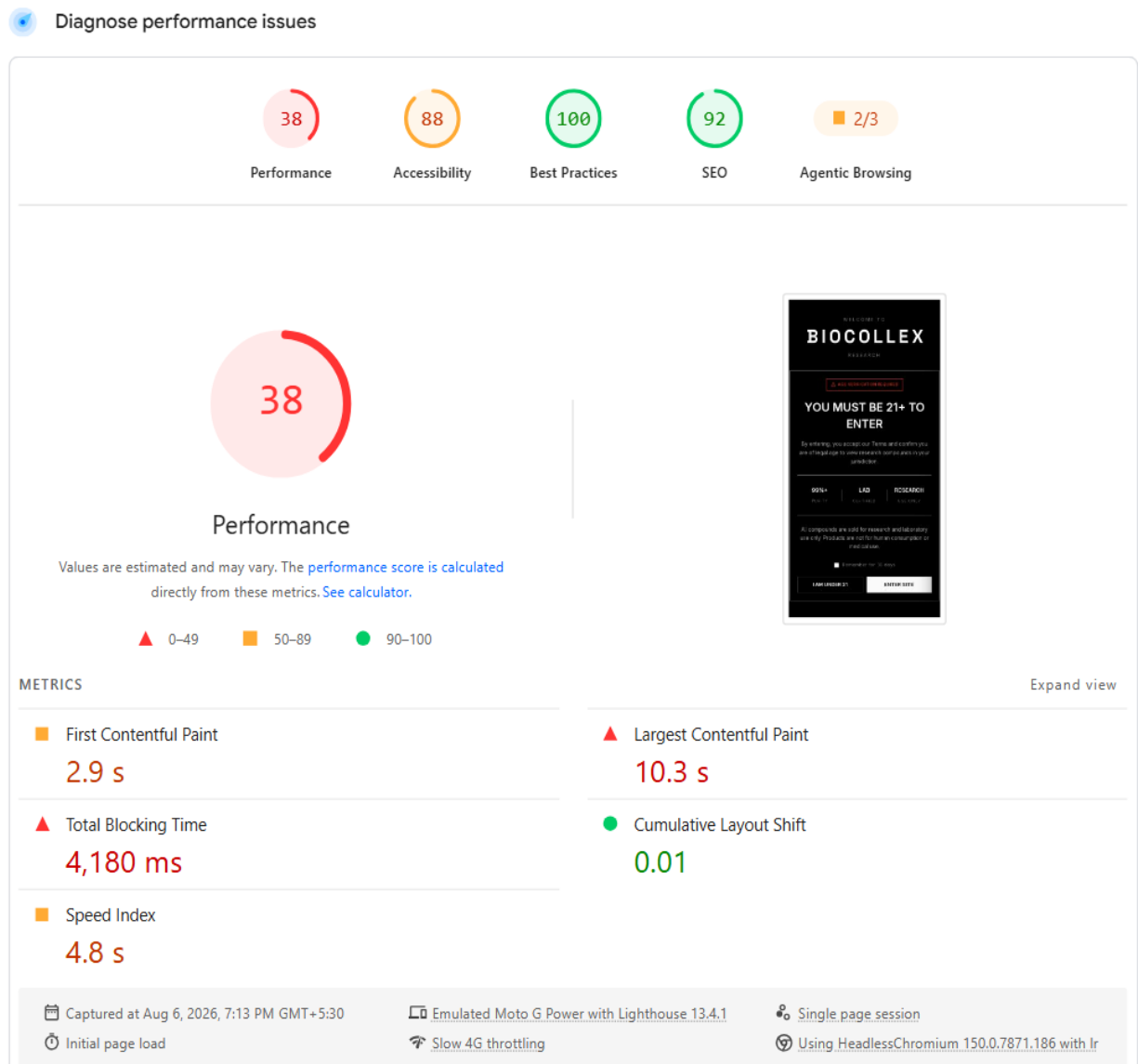
Before vs After

| Metric      | Original Website | Rebuilt Demo |
|-------------|------------------|--------------|
| Performance | 38/100           | 100/100      |
| FCP         | 2.9 s            | 0.9 s        |
| LCP         | 10.3 s           | 1.2 s        |
| TBT         | 4,180 ms         | 0 ms         |
| CLS         | 0.01             | 0            |
| Speed Index | 4.8 s            | 0.9 s        |

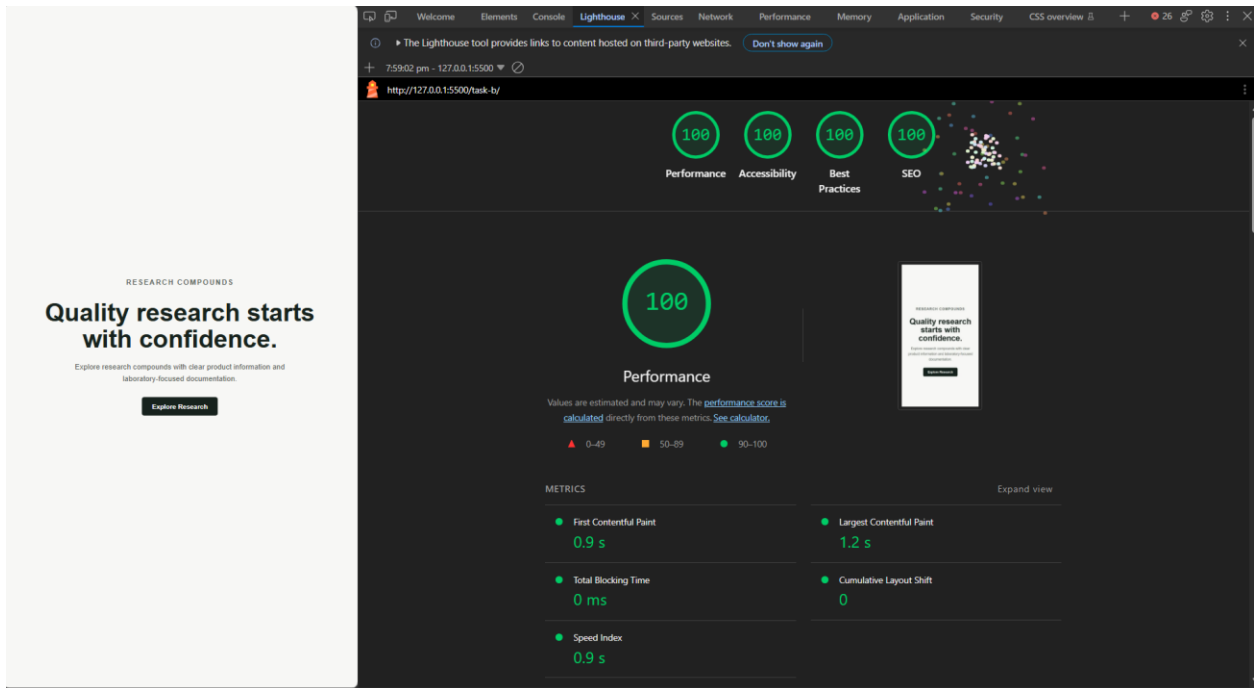
The rebuilt section demonstrates that removing unnecessary heavy resources and keeping the section lightweight can significantly improve loading and responsiveness.

**Note:** The after-results represent the standalone rebuilt demo, not a re-test of the original production website.

## Before



## After



## 6. Client-Facing Summary & Conclusion

- **Client Summary**

The mobile performance review found that the website was carrying more content and browser work than necessary. The biggest contributors were a large hero video, heavy JavaScript, unused CSS and third-party scripts.

The initial Lighthouse performance score was **38/100**, with the main content taking **10.3 seconds to load**.

We rebuilt the hero section using a lightweight approach and **achieved 100/100 Performance** in the standalone demo. The demo also **reduced LCP from 10.3 seconds to 1.2 seconds** and **TBT from 4,180 ms to 0 ms**.

- **Recommended Next Steps**

The highest-value improvements would be:

1. Replace or heavily compress the large hero video.
2. Remove or defer JavaScript that is not required for the initial page.
3. Reduce unused CSS and optimize the main stylesheet.
4. Optimize large images and use modern image formats.
5. Delay non-essential third-party scripts until after the main content loads.

- **Conclusion**

The website does not need a complete redesign to become faster. The largest gains should come from reducing heavy media and unnecessary browser processing.

Based on the demo results, prioritizing these changes should make the website **load faster, become interactive sooner, and provide a better mobile experience,** particularly for users on slower connections or devices.

# Thank you